

Federated Learning with Incomplete Data: When to Use Complete Cases and When to Weight

International Biometric Conference / Seoul, South Korea
July 14, 2026

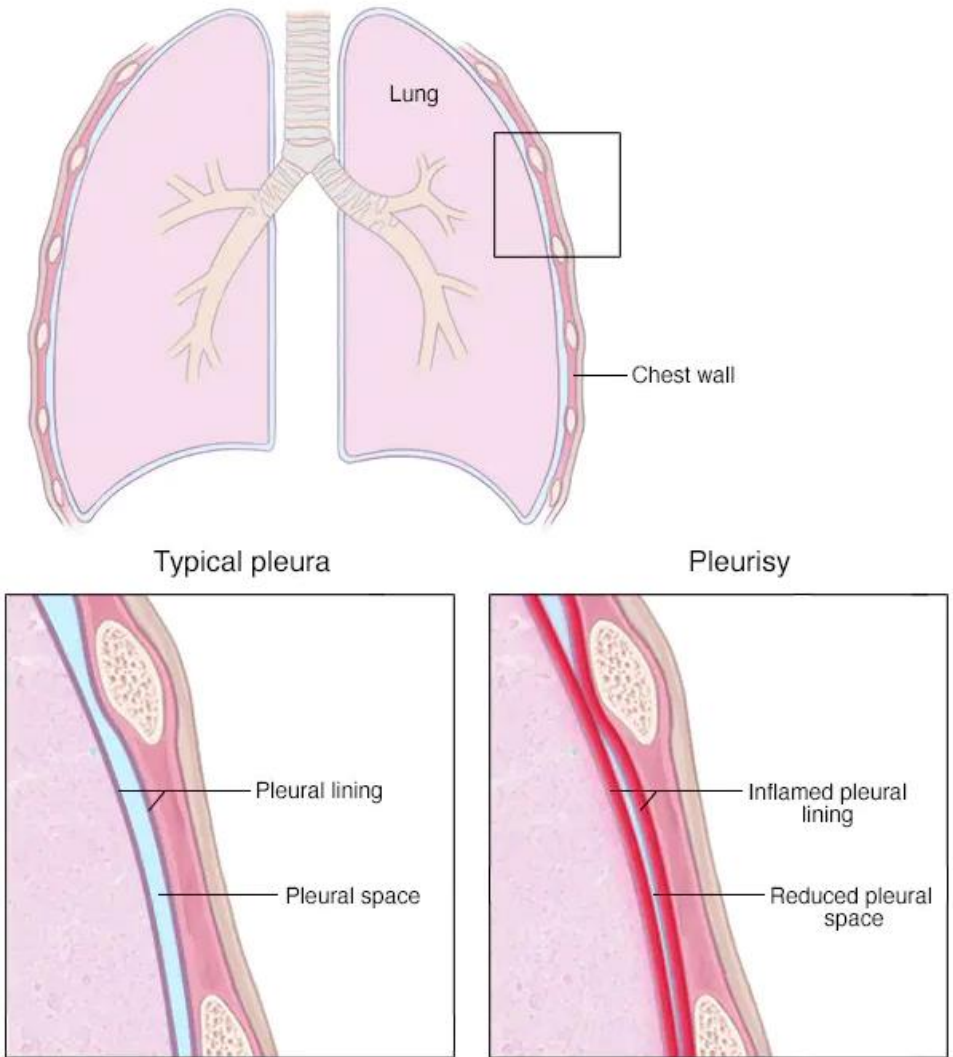
Jesus E. Vazquez, PhD | JHU Provost Postdoctoral Fellow

Yicheng Sheng, Jason Akulian, Chad Hochberg, Theodore J. Iwashyna, Elizabeth A. Stuart, Jiayi Tong



Pleural infections

- **Remains a serious clinical problem**, with substantial morbidity and mortality despite modern antibiotics and drainage strategies¹.

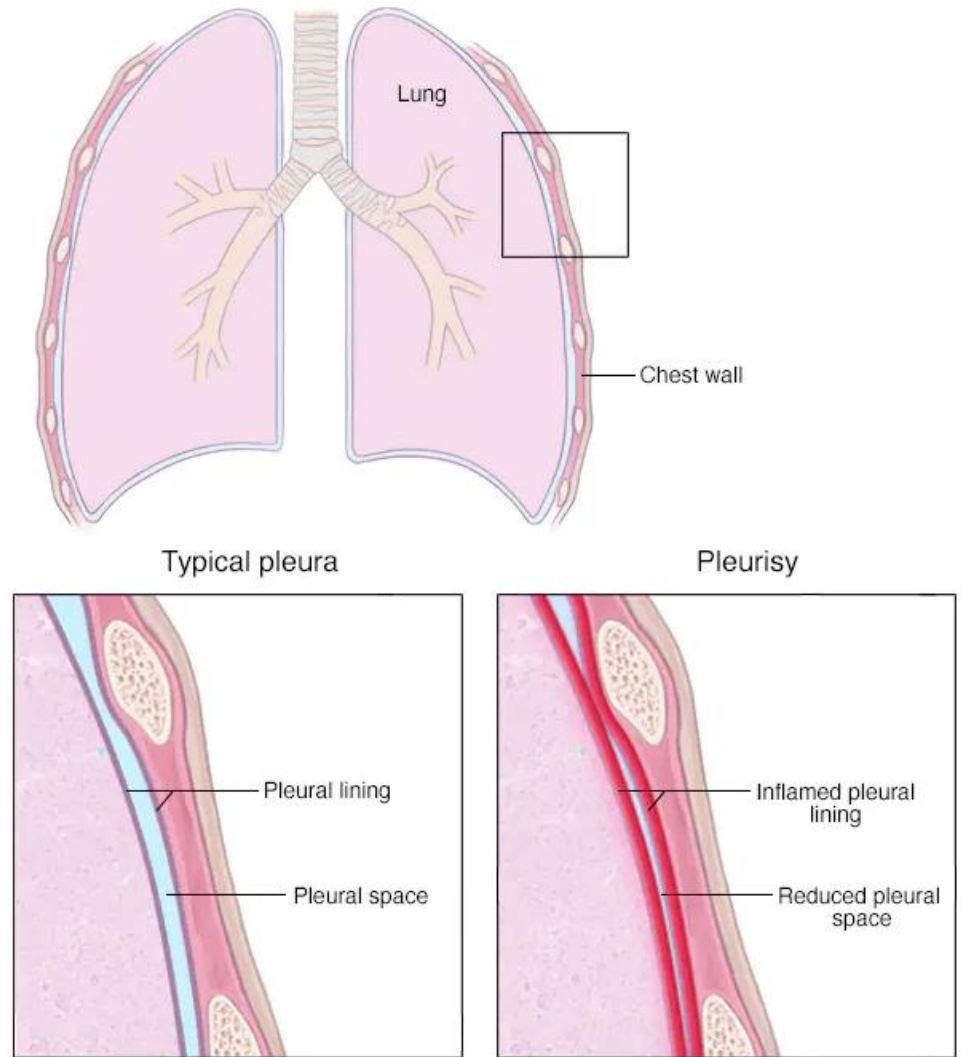


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1. White HD, Henry C, Stock EM, Arroliga AC, Ghamande S. Predicting Long-Term Outcomes in Pleural Infections. RAPID Score for Risk Stratification. Ann Am Thorac Soc. 2015 Sep;12(9):1310-6.

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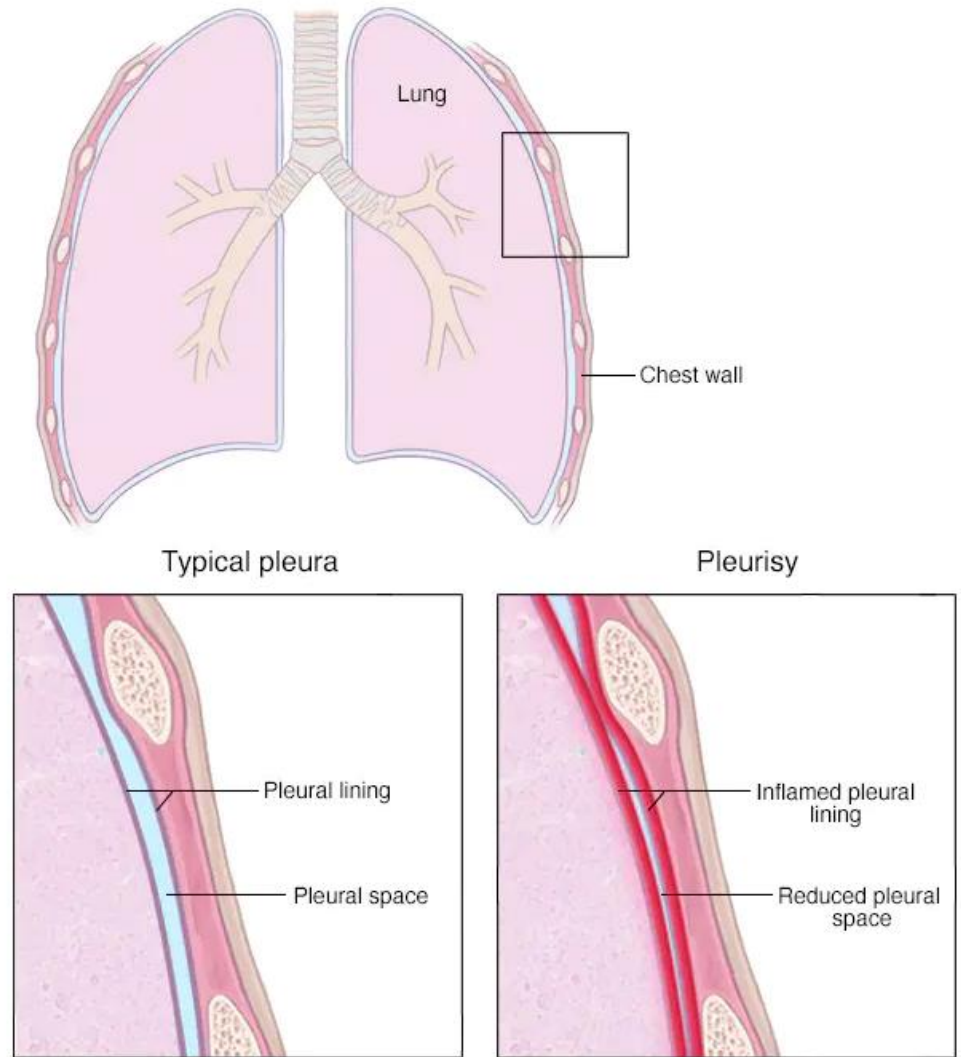


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- **Intrapleural enzyme therapy (IET)** using tissue plasminogen activator and DNase has improved outcomes and reduced the need for surgery, **but treatment response remains highly variable across patients**.
- **Understanding which patients are at highest risk of poor outcomes is important.**



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Data

Two sources: Hopkins (5 Hospitals) and UNC (23 Centers, USA and UK)

	JHU (n=454)	UNC (n=1,494)
90-day mortality	83 (18.3%)	215 (14.4%)
Sex (Male)	344 (75.8%)	1046 (70.0%)
Albumin (g/dL) < 2.7	253 (56.9%)	631 (49.9%)
Missing	10 (2.2%)	228 (15.3%)
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BUN (mg/dL) ≥ 14	180 (39.6%)	500 (33.5%)

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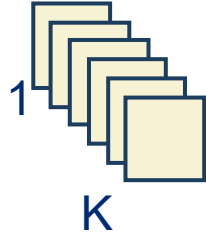
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Questions: How can I replicate a pooled analysis without combining individual level data? How do I adjust for the missing data?

Federated learning vs. missing data



Meta-analysis

 $\hat{\beta}^{(1)}$ $\dots$ $\hat{\beta}^{(K)}$ 

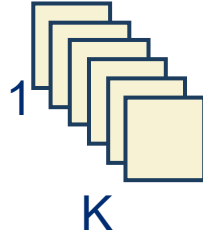
$$\beta_{Meta} = 1/K \times \sum_{k=1}^K \beta^{(k)}$$



Federated learning vs. missing data

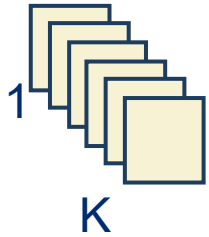


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Federated Learning



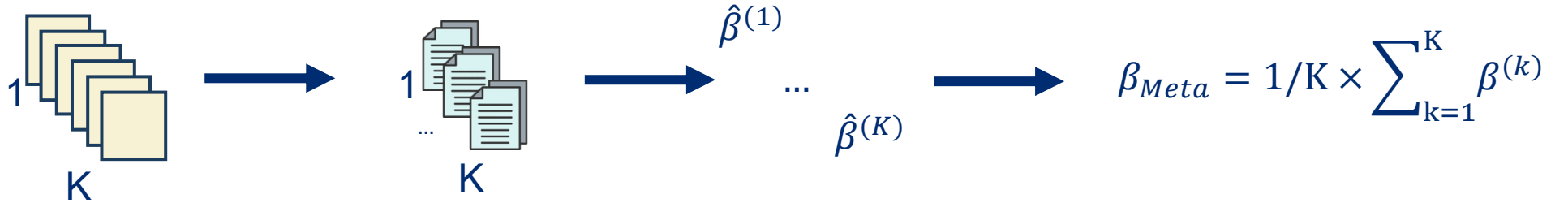
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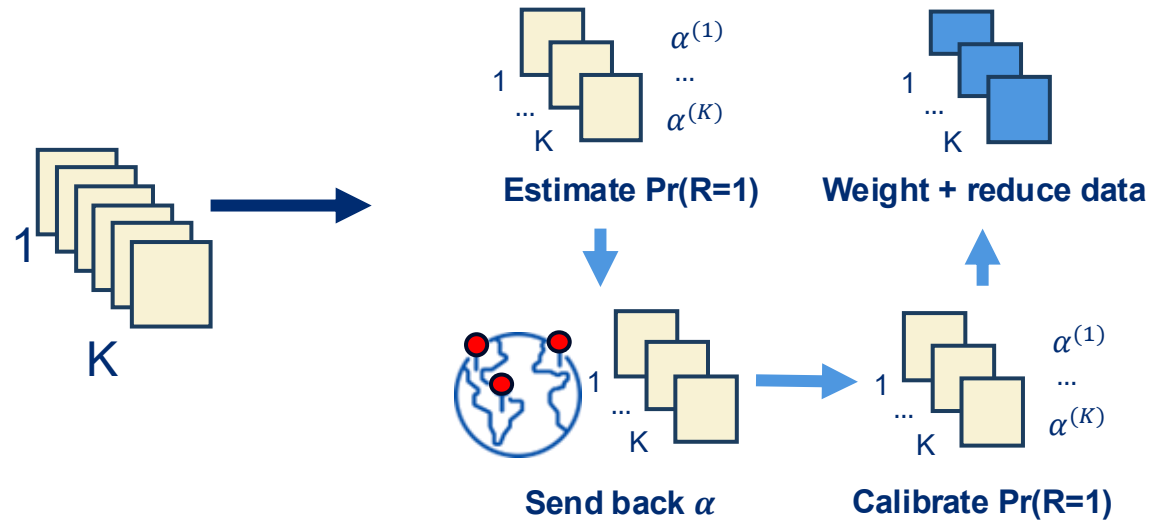
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Federated Learning



Data

Communication Round(s)

The three constraints in federated learning



Privacy

Only summary statistics or
limited information may
leave a site

The three constraints in federated learning



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Communication efficiency

Ideally only few rounds of communication

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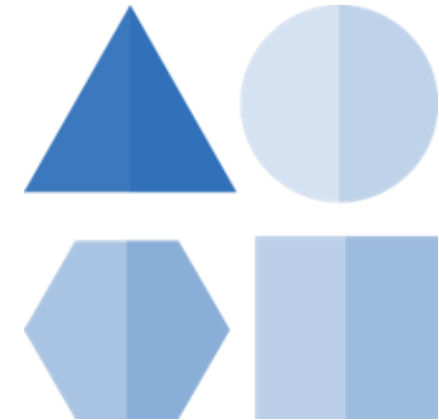
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Heterogeneity

Site-specific distributions and effects

Privacy: What information can I share?

It depends!

- No individual-level data
- Aggregated statistics only
- Data user agreement governs what can be shared



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Centers for Medicare & Medicaid Services (CMM)

- Operates under the US Department of Health and Human Services
- Small cell counts (≤ 10) are suppressed in accordance with CMS disclosure policies to reduce re-identification risk.

Examples: Small Cell Count Suppression

Table 1a. Age Distribution of Patients with Procedure A

Beneficiary age group	Procedure A (N=2,690), N (%)
60-69 years	1,900 (71)
70-74 years	400 (15)
75-79 years	290 (11)
80-84 years	94 (3.5)
85+ years	6 (0.22)



Examples: Small Cell Count Suppression

Table 1a. Age Distribution of Patients with Procedure A

Beneficiary

60-69 years

70-74 years

75-79 years

80-84 years

85+ years

Table 1c. Age distribution of patients who had procedure A

Beneficiary age group	Procedure A (N=2,690), N (%)
60-69 years	>1895 (>70.4)
70-74 years	400 (15)
75-79 years	290 (11)
80-84 years	94 (3.5)
85+ years	<11 (<0.4)



Examples: Small Cell Count Suppression

Table 1a. Age Distribution of Patients with Procedure A

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Table 1b

Beneficiary age

60-69 years

70-74 years

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80-84 years

85+ years

Table 1d. Age Distribution of Patients with Procedure A

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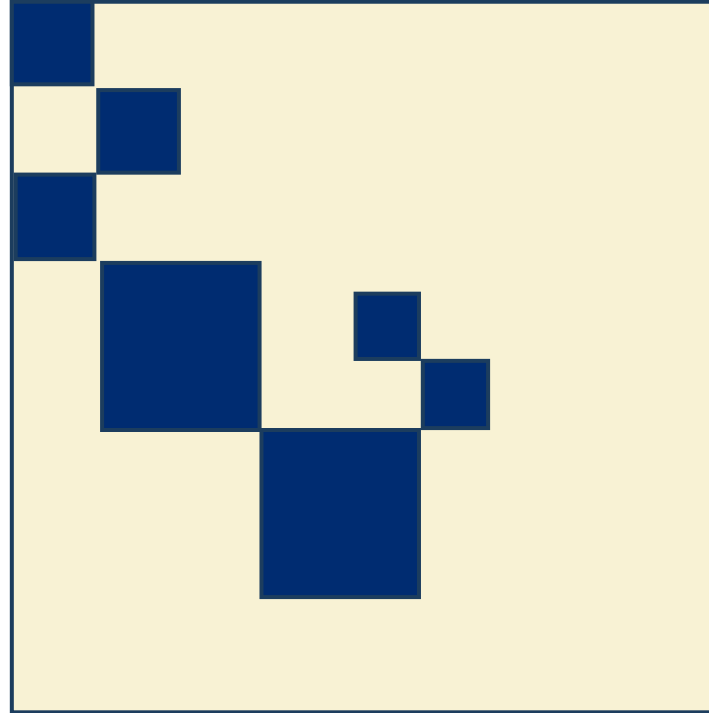
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Answer: Federated learning, however, most assume complete data²

2. Kairouz, P., McMahan, H.B., Avent, B., Bellet, A., Bennis, M., Bhagoji, A.N., Bonawitz, K., Charles, Z., Cormode, G., Cummings, R. and D'Oliveira, R.G., 2021. Advances and open problems in federated learning. Foundations and trends in machine learning, 14(1–2), pp.1-210.

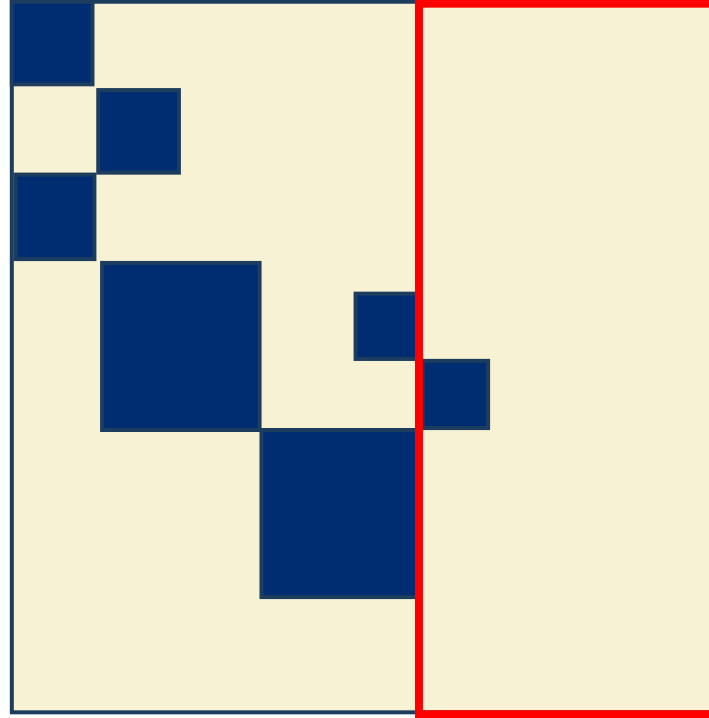
Complete case estimator in federated learning³

The easiest estimator to use:
one restricts the data to only the
observations that have complete
data before applying the model
of interest may only require one
communication between sites.



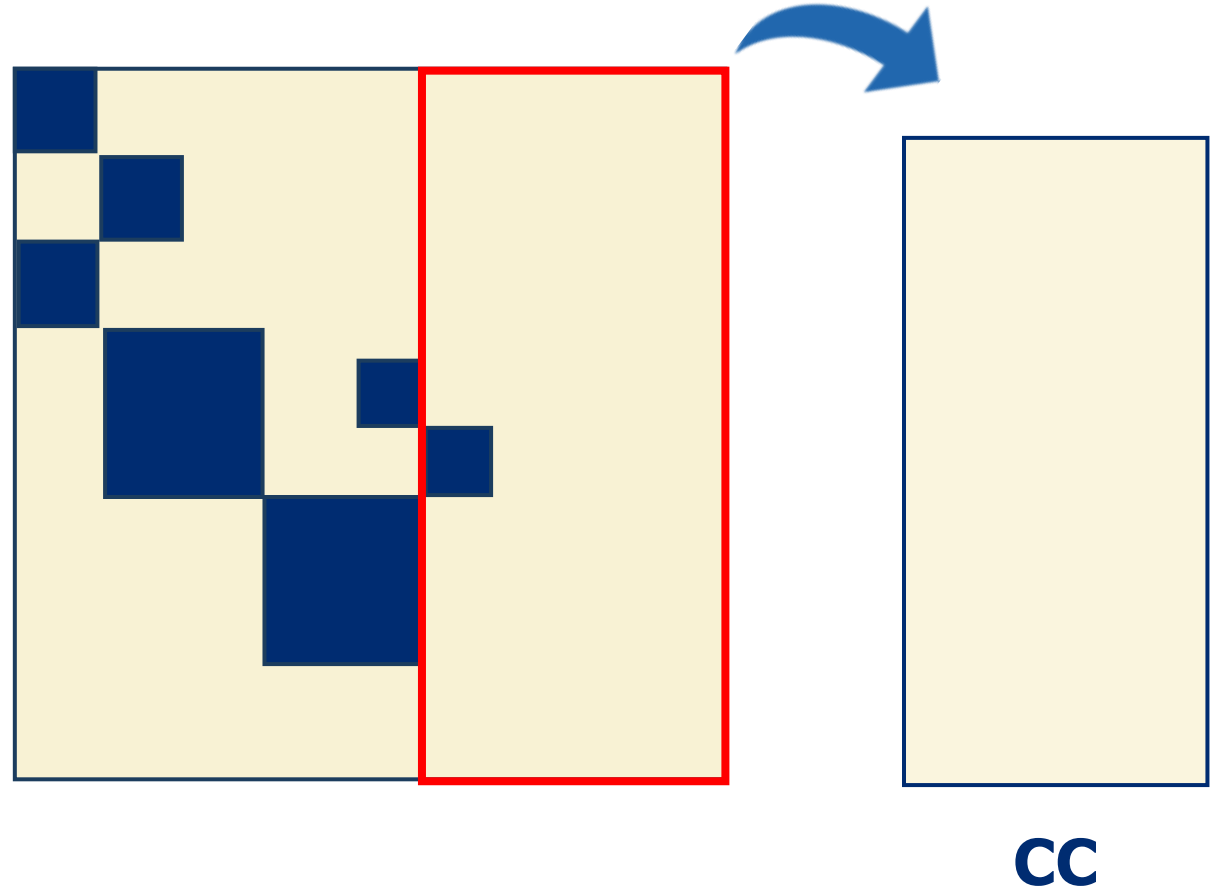
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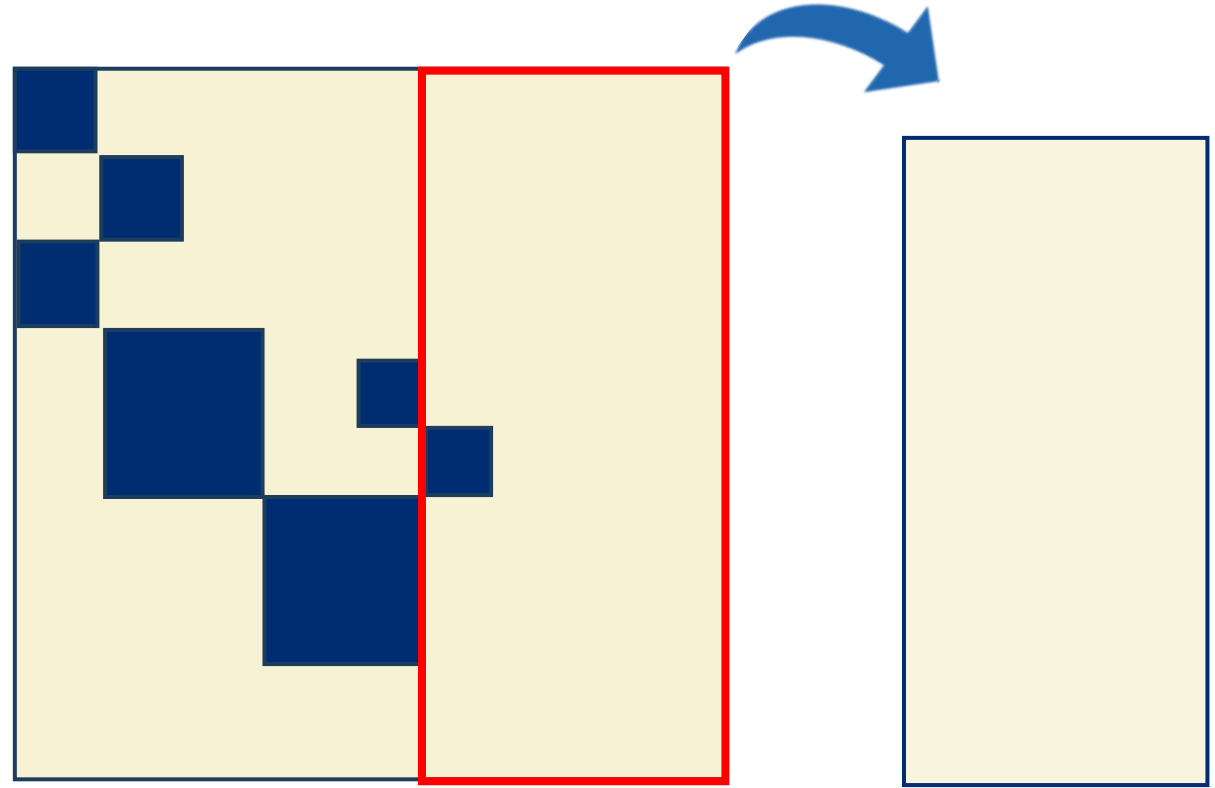
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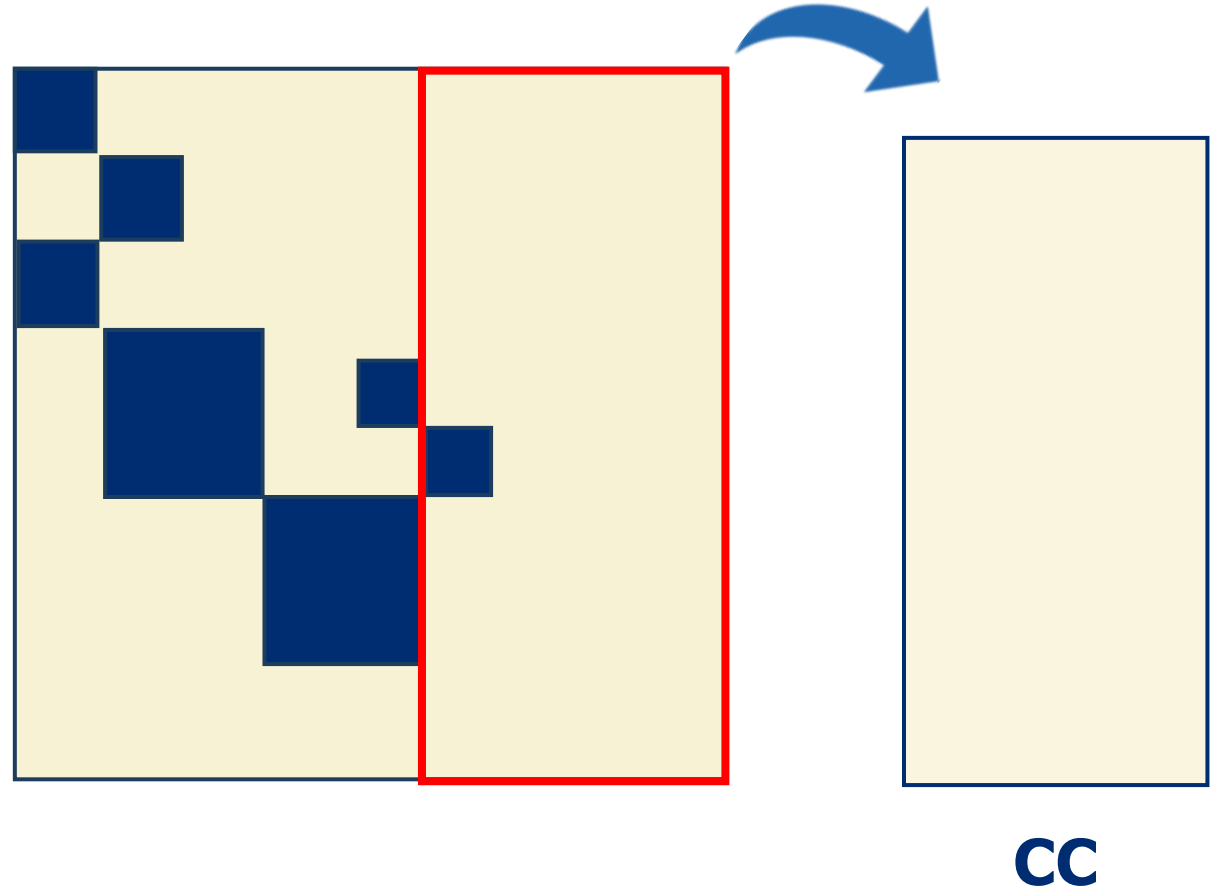
$$\hat{\beta}_{pooled} = \left[\sum_{k=1}^K \{\mathbf{X}^{(k)}, \mathbf{Z}^{(k)}\}^{\top} \{\mathbf{X}^{(k)}, \mathbf{Z}^{(k)}\} \right]^{-1} \left[\sum_{k=1}^K \{\mathbf{X}^{(k)}, \mathbf{Z}^{(k)}\}^{\top} Y^{(k)} \right]$$

CC

3. Chang, Changge, et al. "Multiple imputation for analysis of incomplete data in distributed health data networks." *Nature Communications* 11.1 (2020): 5467.

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**Bias depends on the missingness
mechanism and % of missing
observations**

When is the CC consistent?

	Y	X	(Y, X)
MCAR	$\Pr(R = 1)$	$\Pr(R = 1)$	$\Pr(R = 1)$
MAR	$\Pr(R = 1 \mid X)$	$\Pr(R = 1 \mid Y)$	$\Pr(R = 1 \mid \mathbf{Z})$
	$\Pr(R = 1 \mid \mathbf{Z})$	$\Pr(R = 1 \mid \mathbf{Z})$	
	$\Pr(R = 1 \mid X, \mathbf{Z})$	$\Pr(R = 1 \mid Y, \mathbf{Z})$	
MNAR	$\Pr(R = 1 \mid Y)$	$\Pr(R = 1 \mid X)$	$\Pr(R = 1 \mid Y, X)$
	$\Pr(R = 1 \mid Y, X)$	$\Pr(R = 1 \mid Y, X)$	$\Pr(R = 1 \mid Y, X, \mathbf{Z})$
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			$\Pr(R = 1 \mid X)$
			$\Pr(R = 1 \mid Y)$

Table 2: Probability of complete observation ($R=1$) under different missingness mechanisms, for missingness in Y , X , or jointly (Y, X) . *Cells shaded in gray are indicative that in those scenarios the complete case estimator is consistent.*

General framework

Consider the following regression model

$$E_{Y|X,Z}\{g(Y)\} = m(X, Z; \theta)$$



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We are interested in estimating θ and var θ under two main restrictions.

- Full but unobserved data: for $k \in (1, \dots, K)$, let $O^{(k)} = \{Y^{(k)}, X^{(k)}, Z^{(k)}\}$
- Observed data: $O^{(k)} = \{Y^{(k)}, R^{(k)}, R^{(k)}X^{(k)}, Z^{(k)}\}$
 - $R=1$ denotes when X is observed and 0 if not.



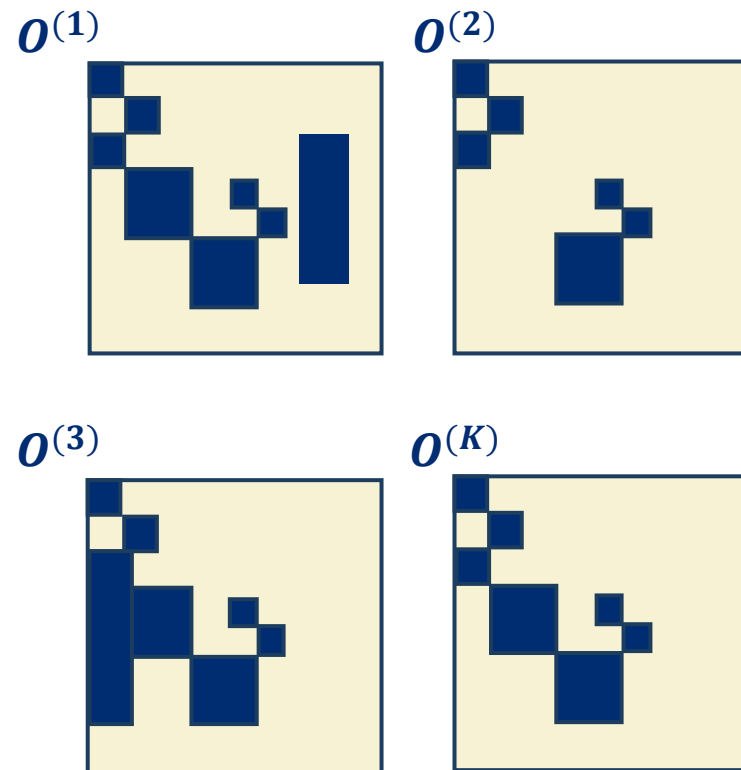
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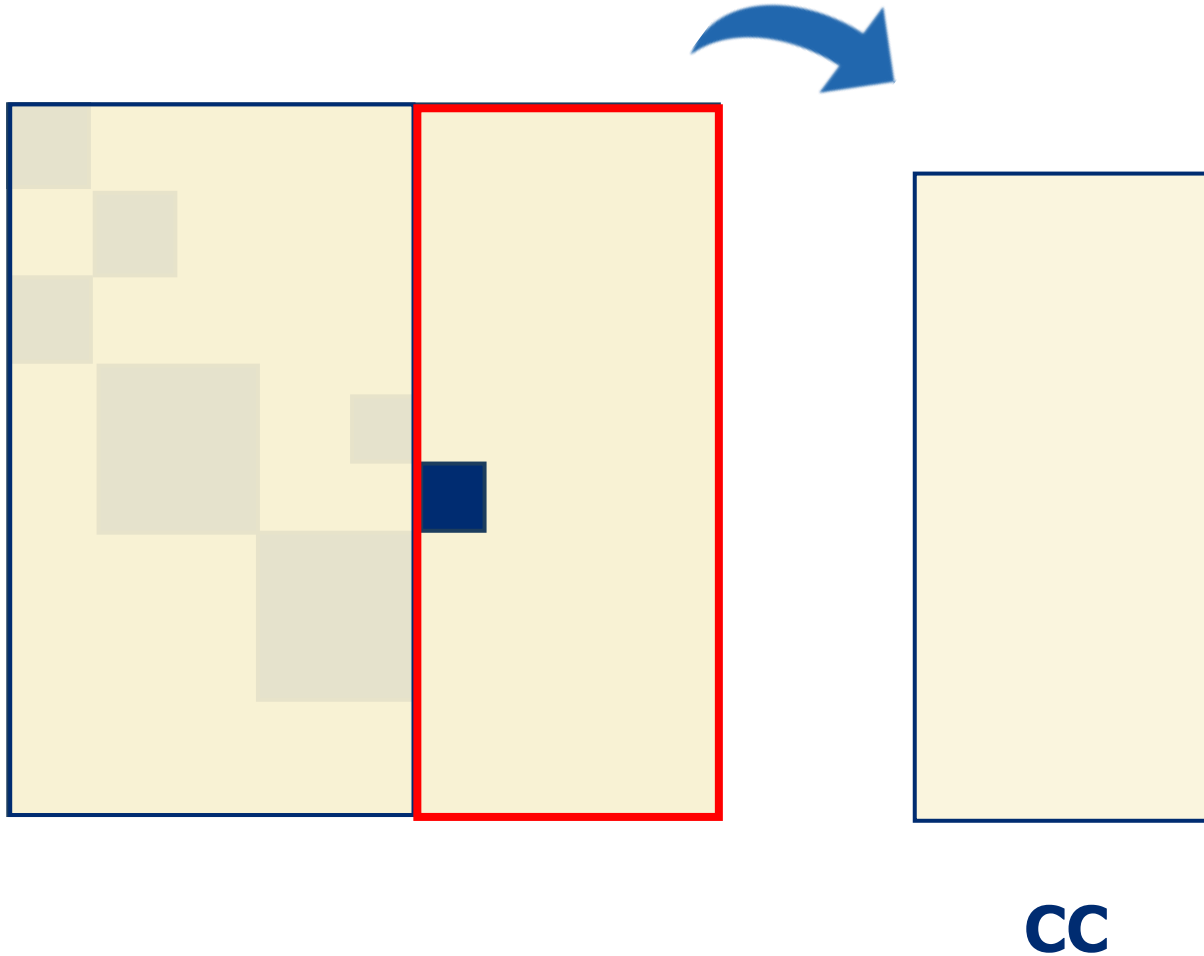
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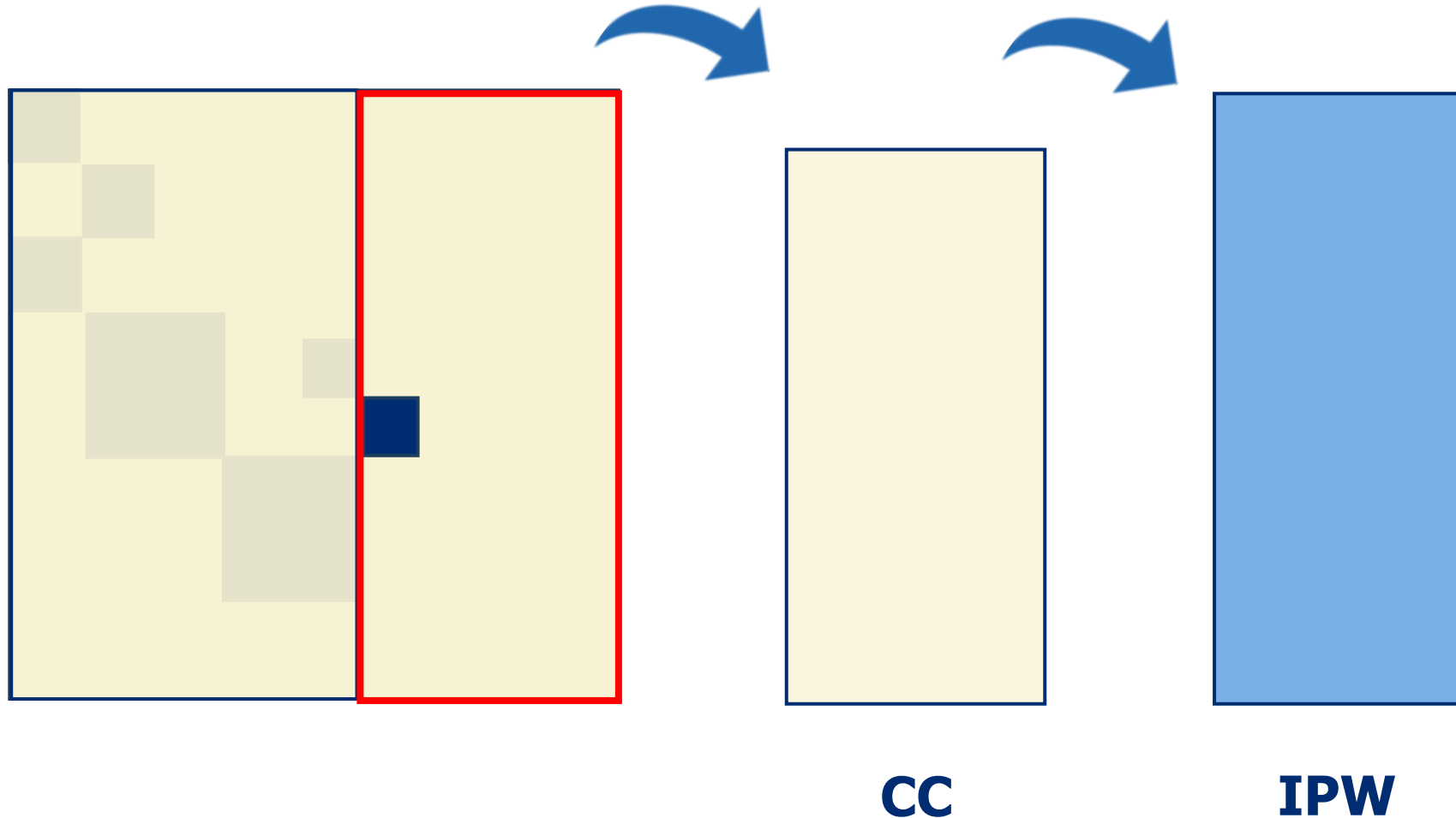
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Inverse probability weighting (IPW) estimator



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Strategy: Weight observations to create a pseudo-sample that approximates the original full data.

M-estimation framework

Let $S(y, x, z; \theta) = \partial \log f_{Y|X,Z}(y, x, z) / \partial \theta^T$. Then, $\hat{\theta}_{IPW}$ is the solution

$$\sum_{i=1}^n \Phi_{IPW}(\mathbf{O}_i; \theta, \alpha) = \sum_{k=1}^K \sum_{n=1}^{n^{(k)}} r_i^{(k)} \frac{\mathbf{S}(y_i^{(k)}, r_i^{(k)} x_i^{(k)}, \mathbf{z}_i^{(k)}; \theta)}{\pi_{Y,X,Z}(y_i^{(k)}, r_i^{(k)} x_i^{(k)}, \mathbf{z}_i^{(k)}; \alpha)} = \mathbf{0},$$

where $\pi_{Y,X,Z}$ is a placeholder for the probability of a complete observation; indexed by a finite dimensional parameter α (e.g., logistic regression model).

Generally, needs correct specification of $\pi_{Y,X,Z}$ (singly-robust) for consistency. Not ideal when sites are different.



Inverse probability weighting estimator with calibrated weights



Multiply robust approach

Multiple guesses for the weights⁴. For example, suppose we have two sites. We will calculate the weight models,

$$\text{Model 1: } g\{Pr(R = 1)\} = \{Y^{(1)}, Z^{(1)}\}\hat{\alpha}^{(1)}$$

$$\text{Model 2: } g\{Pr(R = 1)\} = \{Y^{(2)}, Z^{(2)}\}\hat{\alpha}^{(2)}$$

After J -sites estimates their corresponding $\hat{\alpha}^{(j)}$, these estimates are shared across sites. Let $\alpha = \{\alpha^{(1)}, \dots, \alpha^{(J)}\}$ and $\hat{\alpha}$ be the corresponding estimate, then it follows

$$\gamma_{Y,X,Z}(y_i^{(k)}, r_i^{(k)} x_i^{(k)}, \mathbf{z}_i^{(k)}; \hat{\alpha}) = \left\{ \pi_{Y,X,Z}(y_i^{(k)}, r_i^{(k)} x_i^{(k)}, \mathbf{z}_i^{(k)}; \hat{\alpha}_1), \dots, \pi_{Y,X,Z}(y_i^{(k)}, r_i^{(k)} x_i^{(k)}, \mathbf{z}_i^{(k)}; \hat{\alpha}_J) \right\}^T$$

4. Han, Peisong. "Multiply robust estimation in regression analysis with missing data." *Journal of the American Statistical Association* 109.507 (2014): 1159-1173.

Calibration of the weights

The $\Pr(R=1)$ is then weighted

$$\pi_{Y,X,Z}^*(y_i^{(k)}, r_i^{(k)} x_i^{(k)}, \mathbf{z}_i^{(k)}; \hat{\alpha}) = \hat{\tau}^\top \gamma_{Y,X,Z}(y_i^{(k)}, r_i^{(k)} x_i^{(k)}, \mathbf{z}_i^{(k)}; \hat{\alpha}), \text{ such that}$$

$$\begin{aligned} \gamma_{Y,X,Z}(y_i^{(k)}, r_i^{(k)} x_i^{(k)}, \mathbf{z}_i^{(k)}; \hat{\alpha}) &= \left\{ \pi_{Y,X,Z}(y_i^{(k)}, r_i^{(k)} x_i^{(k)}, \mathbf{z}_i^{(k)}; \hat{\alpha}_1), \dots, \pi_{Y,X,Z}(y_i^{(k)}, r_i^{(k)} x_i^{(k)}, \mathbf{z}_i^{(k)}; \hat{\alpha}_J) \right\}^\top, \\ \hat{\tau} &= \left\{ \sum_{i=1}^{n^{(k)}} \gamma_{Y,X,Z}(y_i^{(k)}, r_i^{(k)} x_i^{(k)}, \mathbf{z}_i^{(k)}; \hat{\alpha}) \gamma_{Y,X,Z}(y_i^{(k)}, r_i^{(k)} x_i^{(k)}, \mathbf{z}_i^{(k)}; \hat{\alpha})^\top \right\}^{-1} \\ &\quad \times \left\{ \sum_{i=1}^{n^{(k)}} \gamma_{Y,X,Z}(y_i^{(k)}, r_i^{(k)} x_i^{(k)}, \mathbf{z}_i^{(k)}; \hat{\alpha}) r_i^{(k)} \right\}. \end{aligned}$$

Estimation of $\Pr(R=1)$

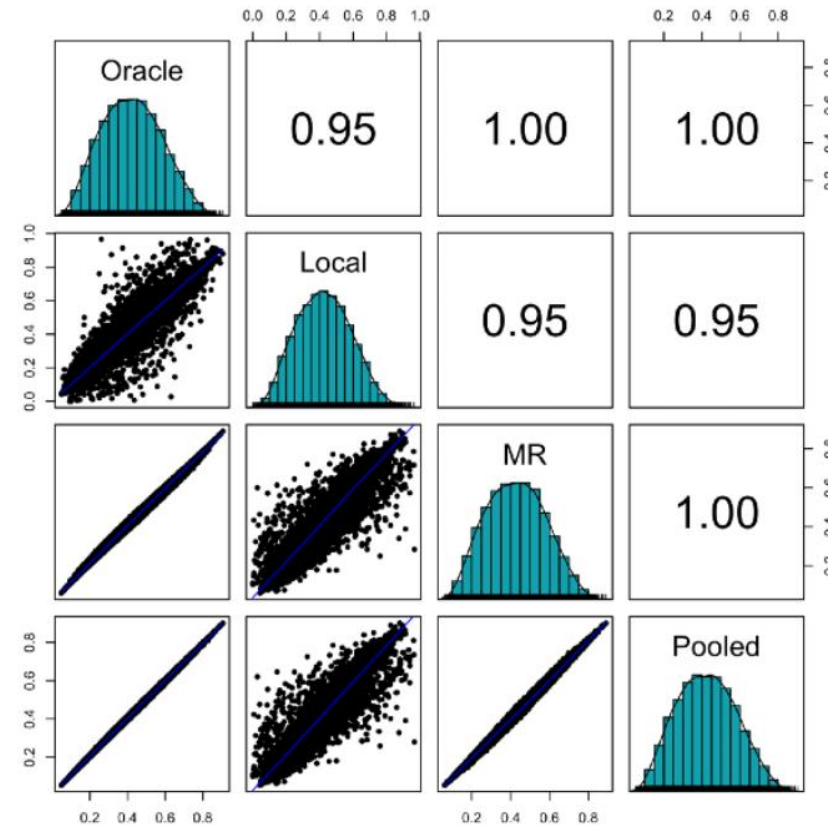


Fig. 3: Scatter plots and Pearson correlation estimates of oracle and estimated probabilities of a complete observation. *Oracle* denotes the true probability of complete observation; *Local* denotes site-specific estimates; *MR* denotes multiply robust estimates obtained by sharing nuisance parameters across sites; and *Pooled* denotes estimates obtained using pooled data. Results shown are for the case of data MAR.

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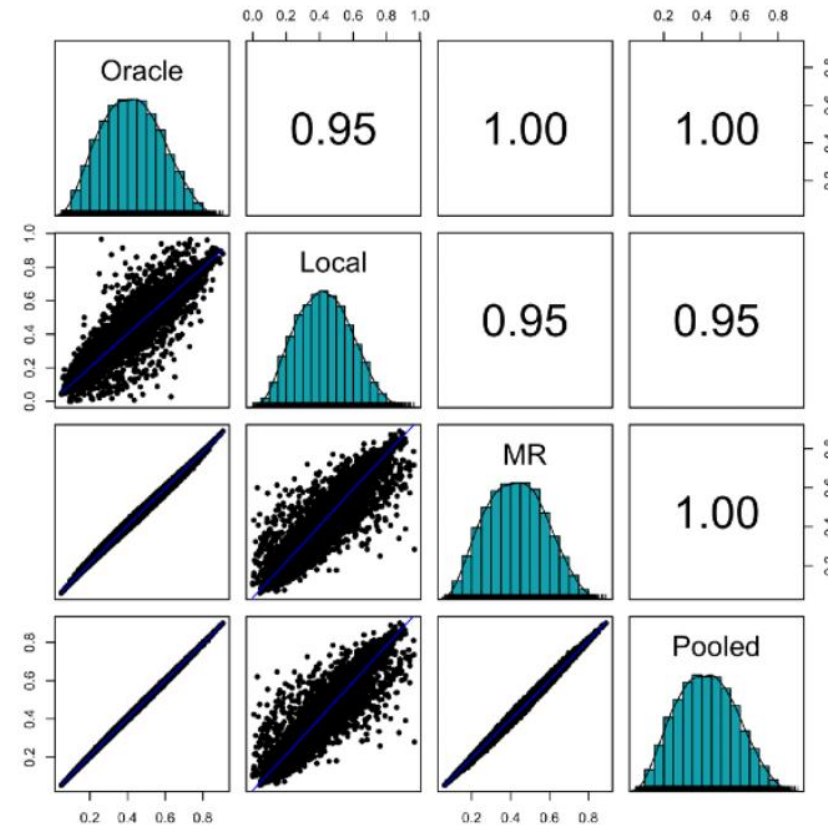


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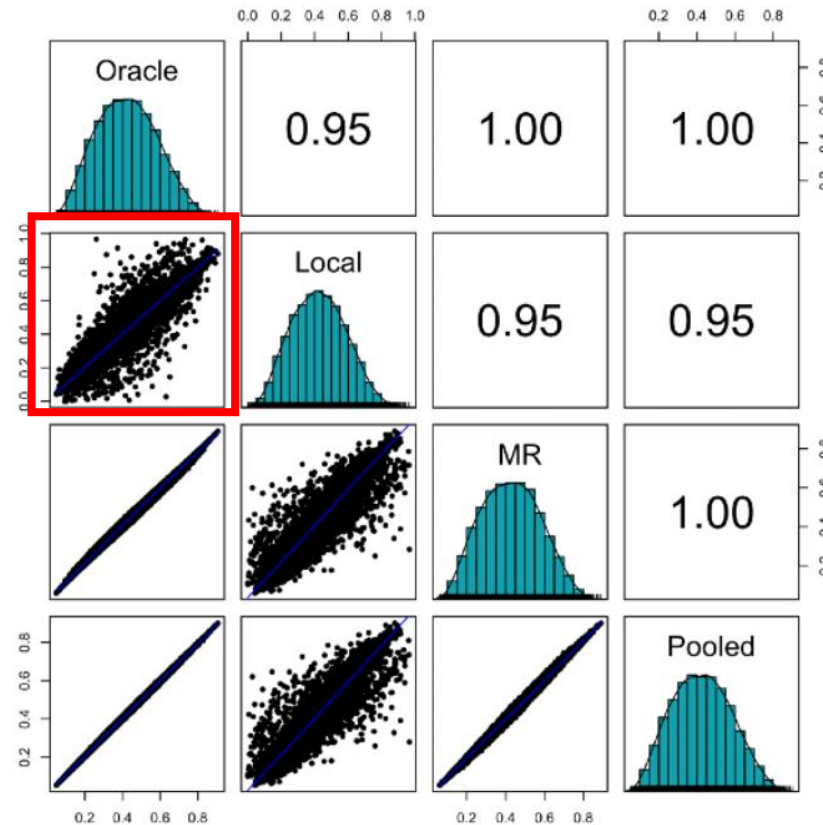


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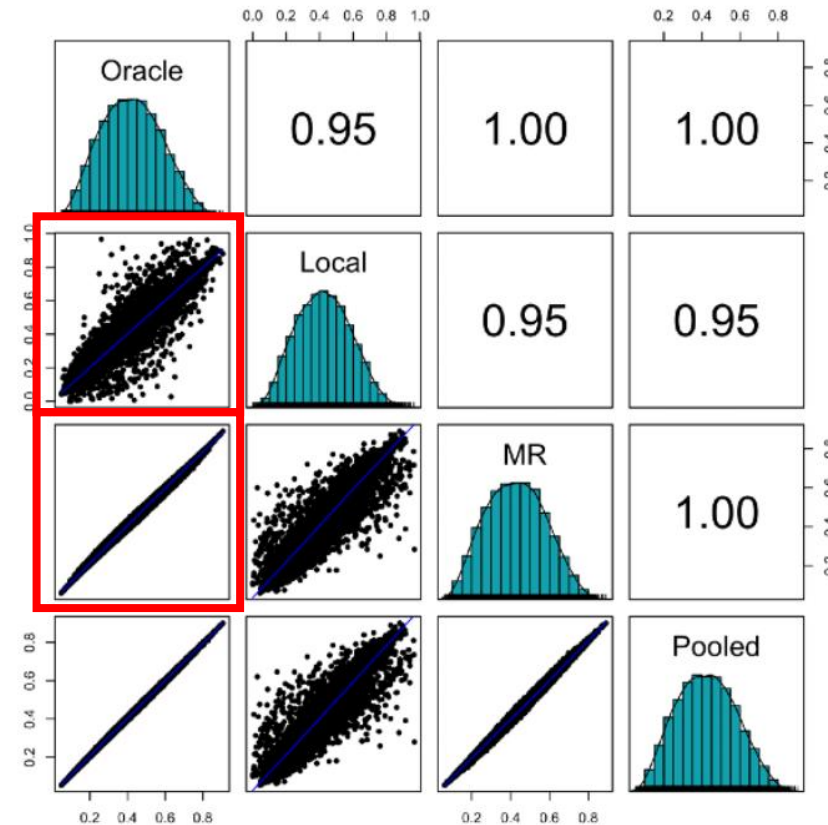


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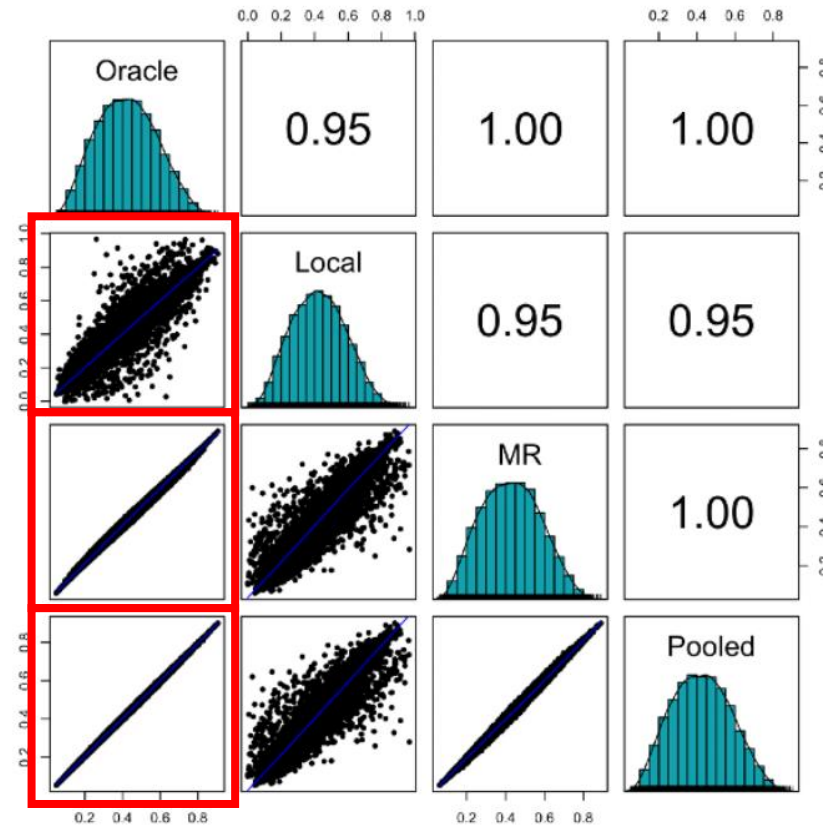


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Robust sandwich estimator

Estimating the missingness model parameters α (and hence the inverse probability weights) introduces uncertainty that can propagate into the outcome model.

Problem: even when θ is consistent, inference can be invalid if the variance is poorly estimated.

Solution: Stacked estimating equation approach⁵. Let $\xi = (\theta^T, \alpha^T, \tau^T)^T$ then we define

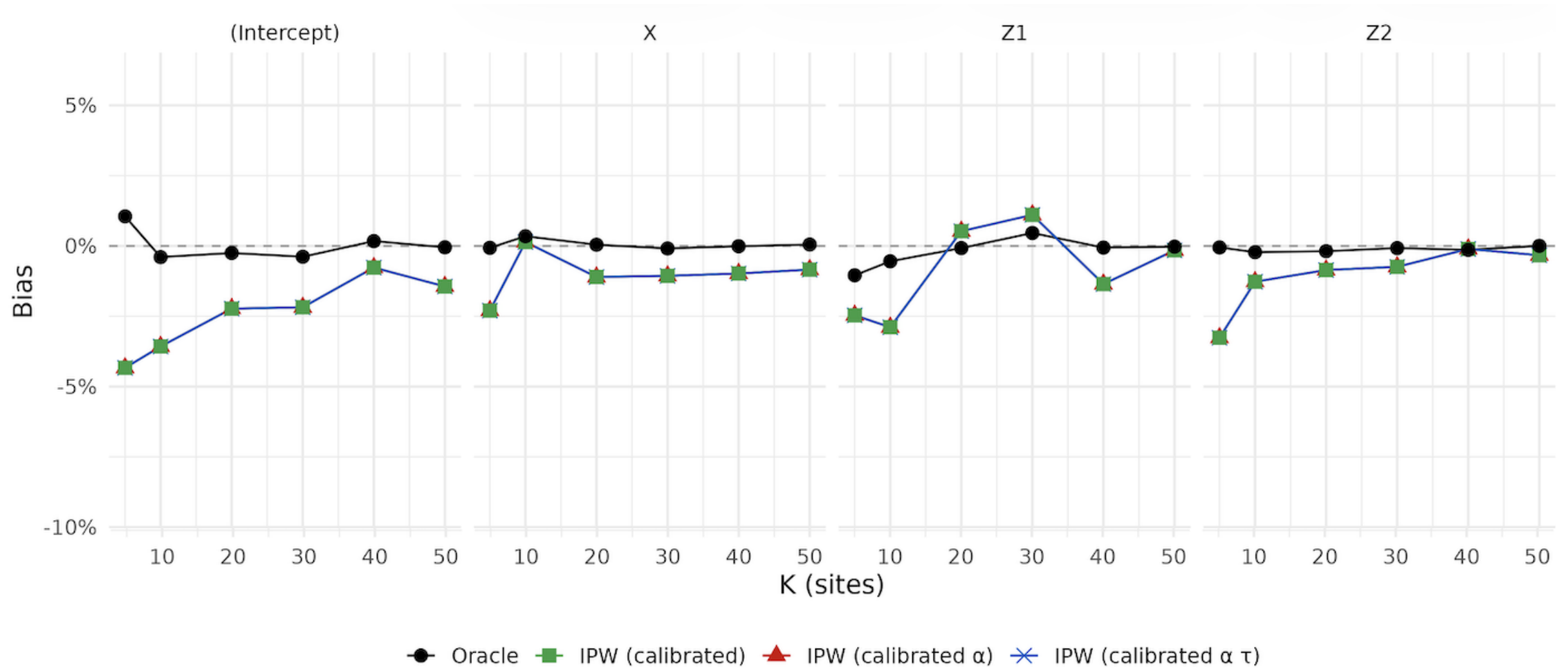
$$\sum_{i=1}^n \Phi_{\text{Stacked}}(\mathbf{O}_i; \xi) = \sum_{i=1}^n \left\{ \begin{array}{c} \Phi_{\text{IPW}}(\mathbf{O}_i; \theta, \alpha, \tau) \\ \Phi_{\text{Site 1, Nuisance}}(\mathbf{O}_i; \alpha_1) \\ \vdots \\ \Phi_{\text{Site J, Nuisance}}(\mathbf{O}_i; \alpha_J) \\ \vdots \\ \Phi_{\text{Site 1, Calibration}}(\mathbf{O}_i; \tau_1, \alpha) \\ \vdots \\ \Phi_{\text{Site K, Calibration}}(\mathbf{O}_i; \tau_K, \alpha) \end{array} \right\} = \sum_{i=1}^n \left\{ \begin{array}{c} \Phi_{\text{IPW}}(\mathbf{O}_i; \theta, \alpha, \tau) \\ \Phi_{\text{Nuisance}}(\mathbf{O}_i; \alpha) \\ \Phi_{\text{Calibration}}(\mathbf{O}_i; \tau, \alpha) \end{array} \right\} = 0$$

5. Tsiatis, Anastasios A. *Semiparametric theory and missing data*. New York, NY: Springer New York, 2006.

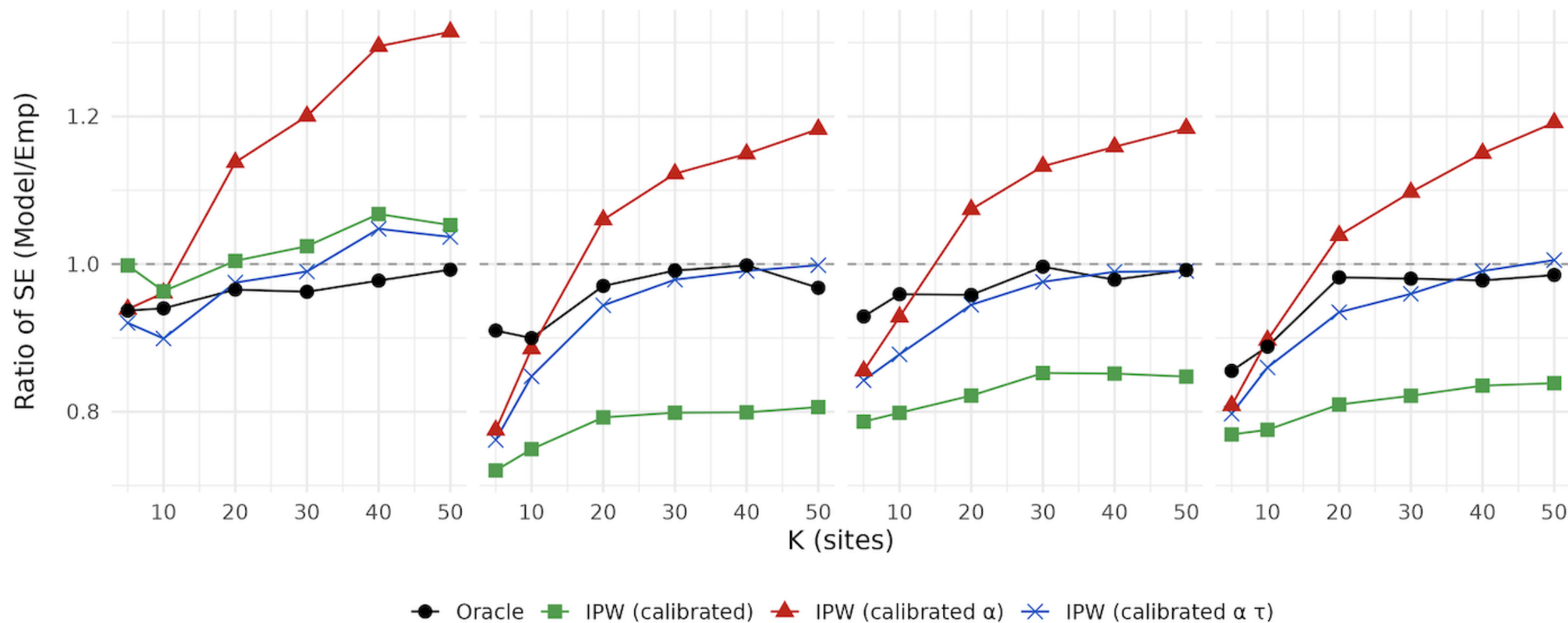
Simulation results



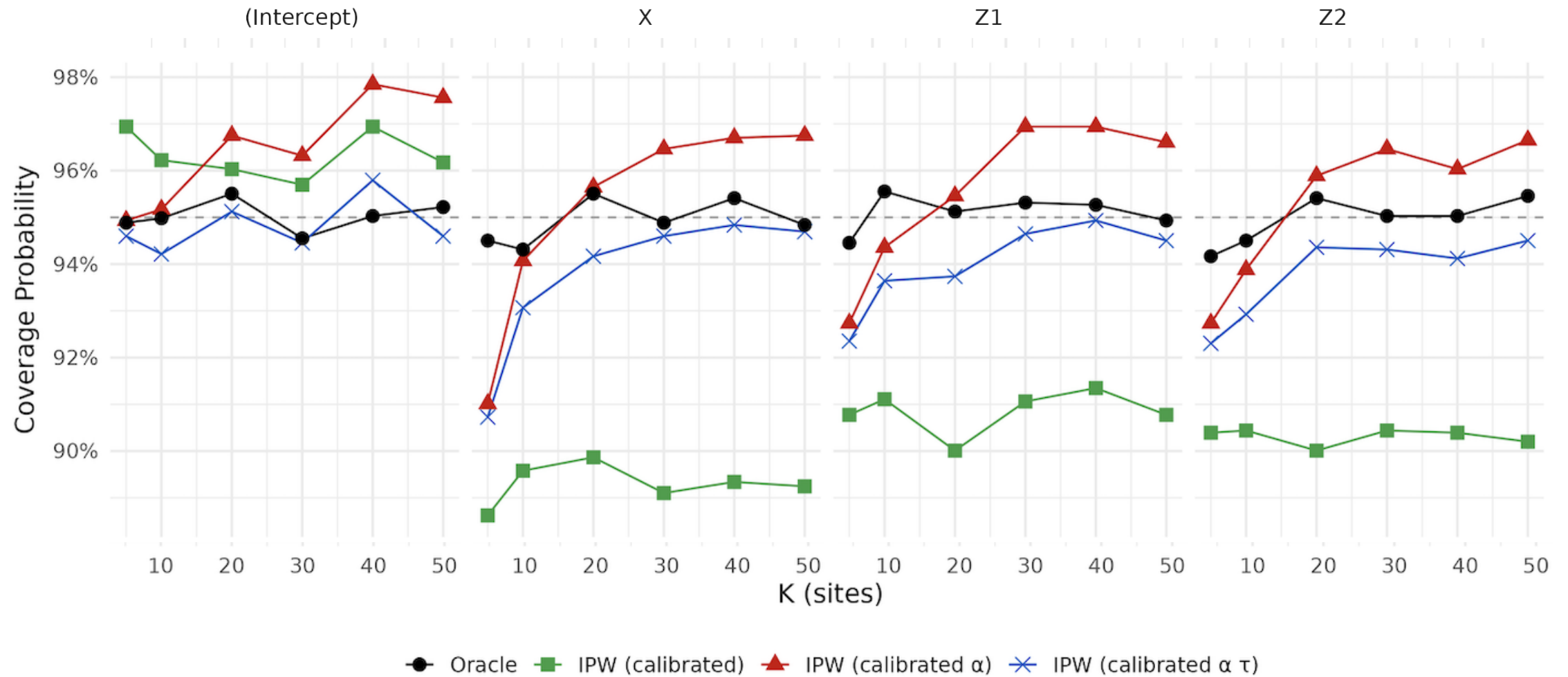
More simulation studies: Bias



More simulation studies: SE ratio



More simulation studies: Coverage



Application to pleural infections

Table 3: Sex-adjusted odds ratio estimate of 90-day mortality (95% Conf. Interval)

	JHU	UNC	Combined
Albumin (g/dL) < 2.7	1.028 (0.770, 1.371) p=0.853, SE=0.216	1.095 (0.768, 1.560) p = 0.618, SE=0.181	1.080 (0.810, 1.441) p = 0.601, SE=0.147

Note: SE reported in log-scale



Application to pleural infections

Table 3: Sex-adjusted odds ratio estimate of 90-day mortality (95% Conf. Interval)

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Federated estimate aligns with site-specific results — The combined OR (1.08) is consistent with both JHU (1.03) and UNC (1.10), demonstrating that federated aggregation recovers similar inference without pooling data.

Efficiency gains from combining sites — The combined confidence interval (0.810, 1.441) is narrower than either site alone.



<https://arxiv.org/abs/2605.20125>

Thank You!

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Works cited

1. White HD, Henry C, Stock EM, Arroliga AC, Ghamande S. Predicting Long-Term Outcomes in Pleural Infections. RAPID Score for Risk Stratification. Ann Am Thorac Soc. 2015 Sep;12(9):1310-6. doi: 10.1513/AnnalsATS.201505-272OC. PMID: 26193196.
2. Kairouz, P., McMahan, H.B., Avent, B., Bellet, A., Bennis, M., Bhagoji, A.N., Bonawitz, K., Charles, Z., Cormode, G., Cummings, R. and D'Oliveira, R.G., 2021. Advances and open problems in federated learning. Foundations and trends® in machine learning, 14(1–2), pp.1-210.
3. Chang, Changgee, et al. "Multiple imputation for analysis of incomplete data in distributed health data networks." Nature communications 11.1 (2020): 5467.
4. Han, Peisong. "Multiply robust estimation in regression analysis with missing data." *Journal of the American Statistical Association* 109.507 (2014): 1159-1173.
5. Tsiatis, Anastasios A. *Semiparametric theory and missing data*. New York, NY: Springer New York, 2006.

